

Letters to the Editor

Domain-Specific LLMs in Radiology: Considerations for Cross-Site and Temporal Robustness

From

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Editor:

We read with great interest the recent article by Dr Zheng and colleagues in the August 2025 issue of *Radiology* (1). This large-scale investigation, based on nearly one million radiology reports, provides important evidence that domain-specific adaptation can enhance the performance of large language models (LLMs) in clinical reporting. The authors demonstrated that their specialized model (LLM-RadSum) outperformed GPT-4o (OpenAI) across factual consistency, coherence, safety, and clinical usefulness, supporting the development of task-focused artificial intelligence tools in radiology.

Several aspects of the work invite clarification and extension. In Table 2 in Zheng et al (1), impression length is presented in words, whereas the subgroup analysis text refers to “impression lengths of 100 characters or more.” It may be helpful to clarify whether these units were intended to align, as such context could further guide readers toward a clearer understanding of the results. External test performance (median F1 of 0.44) was notably lower than internal performance (0.75), reflecting interinstitutional variation in reporting style and terminology. Strategies such as lightweight domain adaptation, harmonized reporting templates, or site-aware prompting could mitigate these differences and improve cross-site robustness.

Follow-up reports were excluded due to incomplete impressions, yet longitudinal comparisons remain integral to radiology practice, particularly in oncology and chronic disease monitoring. Extending LLM-RadSum to incorporate temporal context from prior reports and prior imaging in which feasible would increase clinical relevance. Human evaluation by senior radiologists and oncologists ensured rigor, but expanding assessments to include trainees, referring physicians, and multidisciplinary team members could provide broader perspectives on usability. Reporting interrater agreement and highlighting operational end points (such as edit distance to sign-off or time-to-finalization) in future studies would further demonstrate workflow impact.

Previous research has similarly found advantages for domain-adapted models (2–4). Given the rapid evolution of foundation models, continuous benchmarking against updated general-purpose systems and transparent reporting of methodology will be essential to ensure sustained clinical relevance. By clarifying presentation details and extending evaluation across institutions, longitudinal settings, and diverse user groups, the strong foundation laid by Dr Zheng and colleagues can be translated into safe and effective clinical deployment.

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References

1. Zheng S, Zhao N, Wang J, et al. Comparison of a Specialized Large Language Model with GPT-4o for CT and MRI Radiology Report Summarization. *Radiology* 2025;316(2):e243774.
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Response

From

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We sincerely appreciate the insightful commentary on our study comparing a specialized LLM with GPT-4o for radiology report summarization (1). The feedback from the authors provides valuable perspectives, and we appreciate the opportunity to further clarify and elaborate on aspects of our work.

Regarding the description of impression lengths, the analyses were performed at the character level rather than the word level, because Chinese words are not explicitly separated by spaces. We acknowledge that the term “words” was inappropriately used in Table 2 in our article and confirm that all statistical analyses were conducted at the character level.

In terms of performance, the gap between internal and external test sets reflects the heterogeneity of radiology reporting practices across institutions. We fully agree that strategies such as lightweight domain adaptation, harmonized reporting templates, and site-aware prompting could enhance model generalizability in multicenter settings.

Concerning longitudinal information, the inclusion of follow-up reports is indeed essential for generating radiologic impressions, particularly in oncology and chronic disease monitoring. Our ongoing work involves developing a specialized LLM that integrates findings and impressions from previous reports with current findings to generate the impression for the current examination. By capturing longitudinal textual relationships and report differences, the LLM is expected to produce more comprehensive summaries.

We also agree that incorporating imaging data could further enhance model performance and interpretability by enabling better alignment between textual and visual information. In addition, involving trainees, referring physicians, and multidisciplinary team members in future evaluations could provide valuable insights into how specialized LLMs perform across diverse user scenarios and their practical value in clinical use.

In summary, we remain committed to advancing specialized LLMs for radiology report generation. Future efforts will focus on expanding model applicability through domain adaptation, longitudinal information integration, and multimodal learning. Before clinical deployment, LLMs should be benchmarked and evaluated by different user groups to clarify their strengths and limitations, ultimately facilitating workflow efficiency and

improving diagnostic outcomes. We sincerely appreciate these constructive insights from the authors and look forward to continued progress in this promising field.

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